

THE SOLAR CLOCK

The origin of sundials is lost in ancient history. A stone fragment, dating to about 1500 B.C., on display in a Berlin museum is the earliest known sundial. In the early days of mechanical watches, sundials were often used as a standard for setting. In this modern age of electronic wrist watches sundials have far from vanished into obsolescence. Today these intriguing works of art are primarily used to accent a landscape design, a garden, or a building.

This "garden variety" sundial, though attractive, is difficult to read with reasonable accuracy and requires the use of tables and correction factors. Sundials usually display Local Apparent Time (L.A.T.) where clocks register Standard Time (S.T.). The difference between L.A.T. and S.T. minutes and varies haphazardly over a 30 minute range each year. Several factors account for this difference. Four of these factors produce noticeable error:

1. The Elliptic orbit of the Earth. The angular velocity of the sun as seen from the earth varies throughout the year. The result is a "slow sun" and a "fast sun" twice a year. The error caused by this change ranges from 14 minutes slow to over 16 minutes fast.
2. Longitude. L.A.T. varies directly with longitude. The L.A.T. for Bismark, North Dakota is 68 minutes later than for Vale, Oregon, yet they are both in the Mountain time zone and have the same S.T.
3. Latitude. The latitude of observation determines the average altitude of the semicircular path of the sun. Sunrise and Sunset times are a function of latitude and correction of L.A.T. requires knowledge of the latitude.
4. Refraction. The sun's rays are bent or refracted when entering the earth's atmosphere. The extent of apparent error is greatest at dawn and dusk and may exceed one minute.

A sundial corrected for the above sources of error will provide timekeeping accurate to within a few seconds. The SOLAR CLOCK is such a sundial (The few seconds of error is due to the four-year leap cycle and can be corrected only by using four separate dials, one for each year in the cycle).

The familiar hour line of the common sundial now becomes a distorted figure 8 called an "analemma." The style is exchanged for a "gnomon" (no'-mon), which is a pointed object protruding from the surface of the dial.

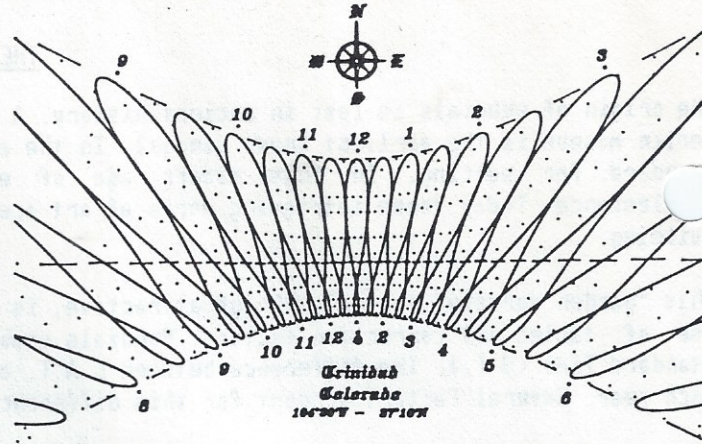
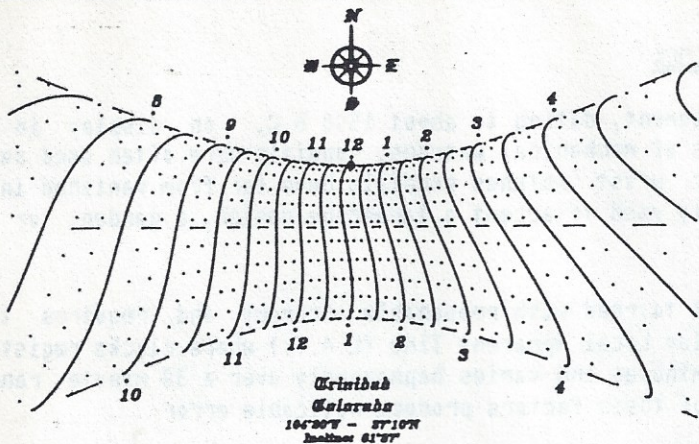
The tip of the gnomon tells the time and also the date. To read the time one must only know the season. During the winter the upper-left quarter of the analemmas is used, spring -- lower-right, summer -- lower-left, and fall -- upper right.

During the day the shadow follows a parabolic path called a day line. Each day of the year has a unique day line. For solar equinox' the line is straight. In the fall and winter they curve up. In the summer and spring they curve down. The height of the gnomon varies with the latitude, averaging 1/10th of the dial height in temperate latitudes.

The standard version of the SOLAR CLOCK has one analemma for each half hour and one day line for the 1st and 16th of each month (payday). A dayline is also included for the solar equinox' and solstices: March 21, Sept. 23, Dec. 21 and June 21. Each day line is actually a series of dots -- one for every ten minutes. Both standard and daylight time are included on the face.

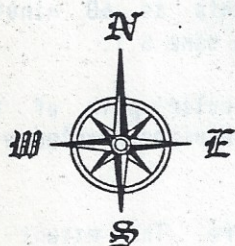
SOLAR CLOCKS come in several flavors and sizes: Paper - for direct use or as a pattern; Transparent sheet - for direct use or for some types of photographic reproduction; and Etched Copper - for an elegant and durable timepiece.

Your SOLAR CLOCK can be created to the size best suited for your application. The face can be customized to be sloped anywhere of from 0 to 90 degrees. No "impossible" hours are included on SOLAR CLOCKS, i.e. hours before sunrise and after sunset are not represented. On the standard SOLAR CLOCK the height of the face is 3/4ths of the width. The Portable model, plotted on an index card, includes a collapsible gnomon and can be used as a compass which indicates true north. Included with each SOLAR CLOCK is a list of information about the particular face, such as the size and location of the gnomon, and complete installation and operating instructions.



OPTION "1" SLOPE "W"

PRICES for SOLAR CLOCKS



WIDTH	PAPER	TRANSPARENT	ETCHED COPPER
1190mm or less	\$12.95	---	---
400mm or less	\$8.95	---	\$59.95 (12in)
255mm or less	\$6.95	\$7.95	\$49.95 (9in)
150mm or less	\$4.95	\$5.50	\$39.95 (6in)
Portable (6in)	\$4.95	---	---

Colorado

ORDER FORM for the standard SOLAR CLOCK

For LOCALE indicate city and state (or anything) to be centered on the lower part of the face. For OPTION enter "P" for Portable, "1" or "2" for first or second half-year only (for eased readability). "S" for standard time only, "C" for cross hatch over face for aid in transfer of pattern. "R" for roman numeral hour labels. "M" for mirror image - reflected across vertical axis. Up to four of these options can be included at no charge. WIDTH is in millimeters. SLOPE is in degrees from horizontal. To set slope parallel to horizontal on the Equator use "E"; Tropic of Capricorn - "S"; Tropic of Cancer - "W". For TIME ZONE print the first letter of your zone, eg "E" for Eastern.

LOCALE: _____ LATITUDE: _____' OPTION: _____ WIDTH/SLOPE: _____ MM _____
 " " _____ LONGITUDE: _____' TIME ZONE: _____

PRINT OR PLACE ADDRESS LABEL BELOW

- ☐ Enclose information on other styles & options.
 (variations of the standard SOLAR CLOCK, glass, granite, and custom faces, custom day lines, line colors & widths, time increments, noon marks.
- ☐ Please figure my longitude and latitude (add \$.50) _____
- ☐ Cellophane coating on Portable SOLAR CLOCK (\$.75) _____
- ☐ Copper gnomon for copper SOLAR CLOCK (with hdwr \$6) _____
- ☐ Mailing Tube for SOLAR CLOCKS over 11in. (\$1.00) _____
 (will otherwise be folded)
- ☐ "21st Century Sundials" -- The complete theory of SOLAR CLOCKS. (\$1.95) _____
- ☐ SOLAR CLOCK(s) ☐ Paper ☐ Transparent ☐ Copper _____

Mail check or money order to: * 2wk. Delivery *
 Doug Gennetten * (Copper - 4wk)*
 4425 Goshawk Dr.
 Ft. Collins, CO 80526 TOTAL ENCLOSED _____